

# Blockchain Fundamentals

Blockchain is a distributed ledger technology that records transactions across a network of computers.

Each block contains a cryptographic hash of the previous block, transaction data, and a timestamp.

Consensus mechanisms like Proof of Work and Proof of Stake ensure agreement among network participants.

Smart contracts are self-executing programs stored on the blockchain that automatically enforce agreements.

# Blockchain Applications

Decentralized Finance (DeFi) provides financial services without traditional intermediaries like banks.

Non-fungible tokens (NFTs) represent unique digital assets on the blockchain for art, gaming, and collectibles.

Supply chain management uses blockchain for transparent tracking of goods from origin to consumer.

Blockchain enables decentralized identity management, giving users control over their personal data.